KG 571: Foundations of Programming
Assignment 2

Submission: .ipynb file (Include all code + explanation in a single ipynb file.
Total Tasks: 5 (each 20 points)

For **each task**, students must include:

- A **Markdown explanation (2–4 sentences)** describing the logic.
- The **Python implementation**
- The **program output**

Task 1 — Data Analysis with Lists

Create a list containing **10 numbers representing daily sales**.

Your program must:

- Print all sales values
- Calculate the **total sales**
- Calculate the **average sales**
- Display the **highest and lowest sales**

Task 2 — Conditional Logic: Grade Evaluation

Write a program that asks the user to enter a **numeric score (0–100)**.

Using conditional statements, determine the **letter grade**:

A : 90–100
B : 80–89
C : 70–79
D : 60–69
F : below 60

Also display whether the student **passed or failed**.

Task 3 — Function: Statistical Calculator

Create a function: *calculate_statistics(numbers)*

The function should accept a **list of numbers** and return:

- Minimum value
- Maximum value
- Average value

Call the function using a sample dataset and display the results.

Task 4 — Class Design: Student Record

Create a class named **Student**.

Attributes:

- name
- student_id
- grade

Methods:

- display_info() → prints student information
- is_passed() → returns True if grade ≥ 60

Create **three student objects** and display their information.

Task 5 — Mini Program: Bank Account Simulation

Create a class **BankAccount**.

Attributes:

- account_holder
- balance

Methods:

- deposit(amount)
- withdraw(amount)
- display_balance()

Create an account and simulate **multiple transactions**.

Display the **final balance**.